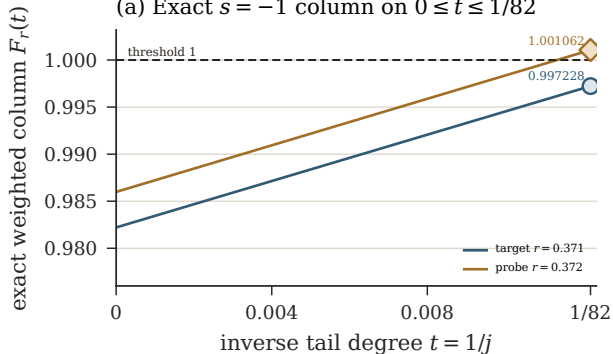




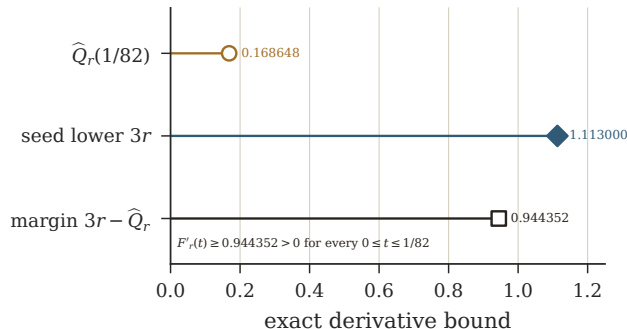
R0.45 fixed negative-charge tail diagnostics

Exact GMP bounds; a derivative theorem replaces the inherited separated estimate for the complete $s = -1$ column

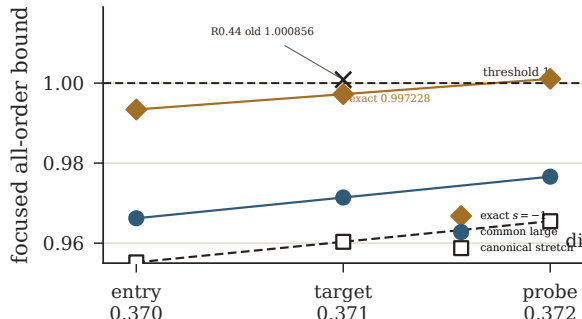
(a) Exact $s = -1$ column on $0 \leq t \leq 1/82$



(b) Derivative lower bound at $r = 0.371$



(c) Consecutive exact radius controls



Theorem: $F'_r(t) \geq 3r - \hat{Q}_r(1/82) > 0$ on $0 \leq t \leq 1/82$; therefore the exact maximum is $j = 82$.

Target: $r = 0.371$ closes at 0.997228; the inherited R0.44 estimate 1.000856 was not the exact induced column.

Control: at $r = 0.372$ the exact $j = 82$, $s = -1$ column is 1.001062, while the common large sector remains 0.976614.

Scope: reduced canonical edge generating system; sampled curves validate presentation, not the all-order theorem.

(d) Complete proof gates at $r = 0.371$

